Name		Date of Performance	
Roll No.		Date of Submission	
Branch		Examined	

Experiment No.

Title: Measurement Of Gear Parameter

Objective:

Measurement of gear parameter viz.

a) Gear tooth thickness.
b) Pitch circle diameter.

Measuring Instruments:

Gear tooth vernier caliper, flanged micrometer, rollers vernier caliper, Gear piece to be tested.

Theory:

The gear transmits power and motion. The motion of the driver gear shall be properly/perfectly uniform with respect to the driven gear only if both gears are of perfect geometrical form and they are mounted perfectly in perfect bearings. Thus, accuracy of gearing is governed by precision in (a) Manufacturing and (b) Measurement of gears.

Gear Terminology:

Module – It is the ration of the pitch circle diameter (pcd) to number of teeth (t).

Pressure Angle – It is the angle made by the normal drawn at the point of contact of mating gears passing through pitch point.

Law of gearing – It states that the common normal at the point of contact always passes through a fixed point i.e. pitch line joining the center of rotation of 2 meshing gears.

Interference – For the correct tooth action the point of contact of two mating teeth must lie on the involute profile. If addendum of one tooth is too large, contact may occur between tip of that tooth and non-involute portion of the mating tooth or in base circle. During analytical inspection of gears the following important gear parameters (elements) are checked.

I) Runout – It is the total variation of the distance between a surface of revolution and a reference surface measured perpendicular to the surface of revolution.

It can be measured and checked by:

Pin check, Rolling check and Contact pattern check.

II) Pitch Index and Tooth to Tooth Spacing – Index refers to the correct angular position of tooth equally spaced about an axis established by a specified accessible proof surface. These elements can be measured by :

Angular positioning device and pitch measuring instrument.

III) Profile – It can be checked by involute profile measuring instrument, profile charts, profile projection.

IV) Lead – It can be checked by instruments and charts.

V) Backlash – It can be defined as the amount by which a tooth space exceeds the thickness of an engaging tooth. It is generally measured at the highest point of mesh on the pitch circle in a direction normal to the tooth surface. It can be measured by dial indicator and a master gear.

VI) Composite – Inspection To Check Eccentricity And For Quality of pitch circle, Wobble etc;

Composite action is the variation in center distance when a gear is rolled in tight mesh. (Double flank contact) with a specified master gear.

It can be measured by,

Gear Rolling Fixtures – The test gear is spring loaded and the variations i.e. composite action recorded. Inspection equipment quality level, statistical sampling plan, interpretation of charts, inspection of (involute) splines and process control.

VII) Tooth Thickness – It can be defined as the length of the arc of a pitch circle corresponding to opposed involutes of a single gear tooth. However for a gear of small pitch, its chordal thickness is approximated as tooth thickness –

We shall be discussing measurement of this element in detail. It can be measured by:

a) Process control
b) Center distance at tight mesh.
c) Backlash at operating center distance
d) Gear tooth vernier caliper.

In addition to the regular horizontal scale it has a second beam at right angles, which carries a tongue. This tongue slides between the jaws of the first main scale. The tongue setting has to be such that when it rests on the top of the tooth, the tips of the jaws at the correct distance down the tooth flank for the required measurement.

i) Chordal thickness at the pitch circle – Let the tongue setting or chordal addendum or the depth of the tooth be 'H'. In this method H=f(t) for some module 'm'. Gears with different number of teeth 'T' will have different 'H', and consequently a different value of 'W' from the fig.

$$H = h + addendum = h + m = R(1-\cos\theta) + m$$

$$\therefore H = \frac{mT}{2}(1-\cos\frac{90}{T}) + m$$

Now,

$$\frac{W}{2} = R\sin\theta = R\sin\left(\frac{360}{4T}\right)$$

$$W = D\sin\left(\frac{90}{T}\right)$$

$$W_{theroitical} = mT\sin\left(\frac{90}{T}\right)$$

With the above 'H', practically tooth thickness $W_{practical}$ is measured and compared with $W_{theorotical}$.

ii) Constant chord Method (W) – The flanks of tooth touches flanks of the base rock. The normals from the point of contact pass through pitch point P and therefore the position and the length of chord AB is fixed and independent of T.

i.e. $H \neq f(T)$ & $W \neq f(T)$

Some module (m) and pressure angle (Ψ)

Here, H=constant

From the figure,

$$H = addendum - CP = m - AP\sin\psi$$

$$H = m - DP\cos\psi\sin\psi$$

$$H = m - \frac{\pi}{4}m\cos\psi\sin\psi$$

$$H = m\left[1 - \frac{\pi}{4}\cos\psi\sin\psi\right]$$

$$Now, AC = (AP)\cos\psi$$

$$= (DP\cos\psi)\cos\psi$$

$$= \frac{circularpitch}{4} \times \cos^2\psi$$

$$= \frac{\pi}{4}m\cos^2\psi$$

$$W_{theroitcal} = \frac{\pi}{4}m\cos^2\psi$$

Thus the chordal addendum tongue setting is not disturbed at all. Here also, W_{pract} is measured and compared with W_{theo}.

e) Plate/Flanged Micrometer or Vernier or David Brown base tangent comparator or span measurement:

Base pitch is defined as the circular pitch of a tooth measured on the base circle (P_b). From the principle of generation of an involutes the distance between opposed involutes or dimension over parallel faces is equal to the distance round the base circle between the points where the corresponding tooth flanks cut i.e. GH=F

$l(A\text{-}B\text{-}C) = l(A_1B_1C_1) = l(A_2B_2C_2) = \text{const.}$

and this value is constant=arc (A_0-B_0-C_0) where, B is the point of tangency between the line AC and arc A_0C_0.

$$Span_{(theoretical)} = mT\cos\psi\left[\tan\psi - \frac{\psi\pi}{180} - \frac{\pi}{2T} + \frac{\pi t_s}{T}\right]$$

Where t_s = number of teeth across which span measurement is considered.

This distance is set in the micrometer with the help of slip gauges.

The different gives the error in tooth thickness at the base circle. Tables are also available which gives 'span' directly for given values of $m_1T_1t_s$.

f) Measurement over pins/Roller – This procedure has been described in detail in the next element to be measured to the tooth thickness; hence tooth thickness can be checked.

VIII) <u>Pitch circle diameter(PCD)</u> - By measurement 'M' –

$$l(DP) = \tfrac{1}{4}PC = \tfrac{1}{4}\times\pi m = \pi m/4$$

$$AP = DP\cos\psi = d/2$$

$$\therefore P_{diameter} = d = \pi m/2\cos\psi$$

d is the diameter of a roller which will rest in tooth space and contact flanks on the pitch circle and lie with its center on pitch circle.

a) If T is even, M=D+d, D=pitch circle diameter.

b) If T is odd,

$$M = 2 \times PQ + d$$

$$M = 2 \times OP \cos\frac{\theta}{2} + d$$

$$\therefore M = D\cos\left(\frac{90}{T}\right) + d$$

$$\therefore D_{act} = \frac{M - d}{\cos\left(90/T\right)}$$

CALCULATIONS:

1) Chordal Thickness Method –

i) Width of tooth -

$$W = Nm\sin\left(\frac{90}{N}\right)$$

$N =$

Minor diameter =

Major diameter =

$D = Nm$

$\therefore m =$

$$W_{th} = Nm\sin\left(\frac{90}{N}\right)$$

$\therefore W_{th} =$

ii) Diameter or Height –

$$d_M = \frac{Nm}{2}\left[1 + \frac{2}{N} - \cos\left(\frac{90}{N}\right)\right]$$